

Assignment 7

1. Multiple Choice Questions (Only one correct answer per question)

1. In a galvanic cell, oxidation occurs at the:
☒ a) Anode
b) Cathode
c) Both electrodes
d) Salt bridge
2. In Zn–Cu cell, electrons flow:
☒ a) $\text{Zn} \rightarrow \text{Cu}$ through external circuit
b) $\text{Cu} \rightarrow \text{Zn}$
c) Through salt bridge
d) In direction of ion diffusion
3. Standard hydrogen electrode (SHE) potential is defined as:
a) +1.00 V
☒ b) 0.00 V
c) -1.00 V
☒ d) +0.76 V
4. Which species is strongest oxidizing agent based on standard potentials?
☒ a) $\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$, $E^\circ = +2.87 \text{ V}$
b) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$, $E^\circ = +1.36 \text{ V}$
c) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$, $E^\circ = +0.80 \text{ V}$
☒ d) $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$, $E^\circ = -0.76 \text{ V}$
5. In Zn–Cu cell with $E^\circ_{\text{cell}} = +1.10 \text{ V}$, ΔG° is:
☒ a) -212 kJ/mol
b) -110 kJ/mol
 $= -1.10 \times 96485 = -1.06 \times 10^5 \text{ J/mol}$

- c) +212 kJ/mol
- d) -55 kJ/mol

6. At 25 °C, $Q < 1$ makes E_{cell} :

- ☒ a) Greater than E°_{cell}
- b) Equal to E°_{cell}
- c) Less than E°_{cell}
- d) Zero

7. In electrolysis of molten NaCl, cathode product is:

- ☒ a) Na
- b) Cl_2
- c) H_2
- d) O_2

8. Faraday's constant $F \approx$:

- ☒ a) $9.6485 \times 10^4 \text{ C mol}^{-1}$
- b) $8.314 \text{ J mol}^{-1} \cdot \text{K}^{-1}$
- c) $0.08206 \text{ L} \cdot \text{atm mol}^{-1} \cdot \text{K}^{-1}$
- d) $6.022 \times 10^{23} \text{ mol}^{-1}$

9. Which statement about corrosion of iron is true?

- a) Cathodic areas oxidize Fe
- b) Anodic areas oxidize Fe^{2+} further
- ☒ c) Oxygen reduction occurs at the cathodic site
- d) Only water needed

10. A concentration cell has same electrodes but [ion] differ. E°_{cell} is:

- ☒ a) Positive
- b) Negative
- ☒ c) Zero
- d) Cannot tell

11. Which occurs at anode in electrolytic cell?

- ☒ a) Oxidation
- b) Reduction
- c) Both
- d) Neither

12. In Nernst eqn, increasing products concentration:
- a) Increases E_{cell}
 - ☒ b) Decreases E_{cell}
 - c) No change
 - d) Equals E°_{cell}
13. The main advantage of fuel cells over combustion engines:
- a) Lower efficiency
 - ☒ b) Convert chemical energy directly to electrical energy
 - c) Use fossil fuel only
 - d) No need for redox reactions
14. Alpha decay emits:
- a) Electron
 - ☒ b) Helium-4 nucleus
 - c) Positron
 - d) Photon
15. In β^{-} decay:
- a) Proton $\rightarrow$ neutron
 - ☒ b) Neutron $\rightarrow$ proton + electron
 - c) Electron capture by proton
 - d) Positron emission
16. β^{+} decay (positron emission) converts:
- ☒ a) Proton $\rightarrow$ neutron + positron
 - b) Neutron $\rightarrow$ proton + electron
 - c) Electron $\rightarrow$ positron
 - d) Proton $\rightarrow$ neutron + electron
17. Electron capture involves:
- ☒ a) Proton capturing orbital electron $\rightarrow$ neutron
 - b) Electron emission from nucleus
 - c) Beta particle emission only
 - d) Gamma emission
18. Gamma rays differ from α and β in that they:
- a) Have mass 0 and charge 0
 - b) Have highest penetration

- c) Usually accompany other nuclear decays
- ☒ d) All of these

19. Which radiation has the least penetration?

- ☒ a) α
- b) β
- c) γ
- d) All equal

20. Stable isotopes with $Z \leq 20$ generally have n/p ratio:

- ☒ a) 1
- b) <1
- c) >1
- d) 0

21. Half-life is:

- a) Time for all nuclei to decay
- ☒ b) Time for half of radioactive nuclei in sample to decay
- c) Time to reach stable state
- d) Proportional to molar mass

22. Units of activity in SI:

- a) Ci
- ☒ b) Bq
- c) Gy
- d) Sv

23. Which instrument detects radiation via ionization of gas?

- ☒ a) Geiger counter
- b) Scintillation counter
- c) Film badge
- d) Mass spectrometer

24. Nuclear fission is:

- a) Combining nuclei
- ☒ b) Splitting heavy nucleus into lighter nuclei
- c) Beta decay
- d) Electron capture

25. Moderator in reactor:

- a) Absorbs neutrons
- ☒ b) Slows neutrons
- c) Shields radiation
- d) Initiates fission

26. Fusion in stars primarily combines:

- ☒ a) H to He
- b) C to O
- c) Fe to Ni
- d) H to Li

27. PET scans use isotopes that emit:

- a) α particles
- ☒ b) β^+ (positrons)
- c) β^- particles
- d) γ from decay chains only

28. Acute high radiation doses cause:

- ☒ a) DNA damage and possible death
- b) Mild heating only
- c) Increased photosynthesis
- d) No biological effect

29. The general formula C_nH_{2n+2} applies to:

- ☒ a) Alkanes
- b) Alkenes
- c) Alkynes
- d) Aromatics

30. The correct IUPAC name for $CH_3CH_2CH_2CH_3$ is:

- a) Methane
- b) Ethane
- ☒ c) Butane
- d) Pentane

31. A carbon-carbon double bond is characteristic of:

- a) Alkynes

- b) Aromatics
- ☒ c) Alkenes
- d) Haloalkanes

32. In cyclohexene, the ring contains:

- ☒ a) One double bond, sp^2 carbons
- b) One triple bond, sp carbons
- c) Only σ bonds
- d) Alternating single and triple bonds

33. Which functional group is $-\text{COOH}$?

- a) Aldehyde
- b) Ketone
- ☒ c) Carboxylic acid
- d) Ester

34. Which has the highest boiling point?

- a) CH_4
- ☒ b) $\text{C}_2\text{H}_5\text{OH}$
- c) CH_3OCH_3
- d) C_3H_8

35. Ketones contain:

- a) $\text{C}=\text{O}$ at end of chain
- ☒ b) $\text{C}=\text{O}$ within chain
- c) $\text{C}-\text{OH}$ group
- d) NH_2 group

36. Which is an example of a structural isomer?

- a) cis-2-butene and trans-2-butene
- ☒ b) Ethanol and dimethyl ether
- c) (R)-2-butanol and (S)-2-butanol
- d) Cyclohexane and benzene

37. Which describes enantiomers?

- a) Same formula, different connectivity
- ☒ b) Same connectivity, non-superimposable mirror images
- c) Differ only in double bond position

d) Differ only in molecular weight

38. The amino group is:

- ☒ a) -NH_2
- b) -COOH
- c) -OH
- d) -C=O

39. Primary structure of protein refers to:

- ☒ a) Sequence of amino acids
- b) α -helix and β -sheet
- c) Overall 3D shape
- d) Aggregation of multiple subunits

40. The bond joining amino acids in proteins is:

- a) Glycosidic
- b) Ester
- ☒ c) Peptide (amide)
- d) Hydrogen

41. DNA differs from RNA by:

- a) Sugar type and base composition
- b) Presence of thymine instead of uracil
- c) Being double-stranded
- ☒ d) All of the above

42. Complementary base pairing in DNA is:

- a) A-G, C-T
- ☒ b) A-T, G-C
- c) A-U, G-C
- d) A-C, G-T

43. Lipids that form bilayers in cell membranes are:

- a) Steroids
- ☒ b) Phospholipids
- c) Triglycerides
- d) Fatty acids

44. In an ester functional group, the C=O carbon is bonded to:

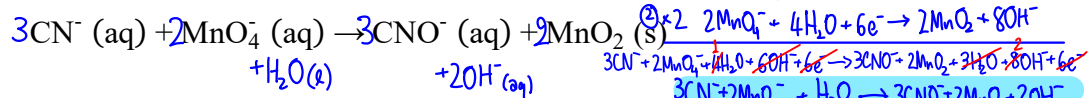
- a) OH only
- b) OR group
- c) NH₂ group
- d) CH group only

45. Which macromolecule carries genetic information?

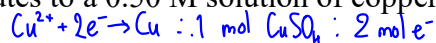
- a) Protein
- b) Polysaccharide
- c) Lipid
- d) Nucleic acid

2. Short-Answer/Calculation Questions

1. Complete and balance this equation for a redox reaction that takes place in basic solution:



2. How many grams of copper will be plated out by a current of 2.3 A applied for 35 minutes to a 0.50 M solution of copper (II) sulfate?



$$\text{Total charges } Q = It = 2.3 \text{ C/s} \times 35 \text{ min} \times 60 \text{ s/min} = 4830 \text{ C}$$

$$\text{Cu mass} = \frac{4830 \text{ C}}{96485 \text{ C/mol e}^-} \times \frac{1 \text{ mol CuSO}_4}{2 \text{ mol e}^-} \times \frac{1 \text{ mol Cu}}{1 \text{ mol CuSO}_4} \times \frac{63.546 \text{ g Cu}}{1 \text{ mol Cu}} = 1.59 \text{ g}$$

3. The concentrations of zinc ion in the two compartments are 4.50 M and 1.11×10^{-2} M, respectively. Calculate the cell emf.

$$E^\circ = 0$$

$$n = 2$$

$$F = 96485 \text{ C/mol e}^-$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

$$= 0 - \frac{8.314 \times 298}{2 \times 96485} \ln \left(\frac{1.11 \times 10^{-2}}{4.50} \right) = 0.077 \text{ V}$$

4. A sample's activity drops to 12.5% in 24 d. Find $t_{1/2}$.

$$12.5\% = \frac{1}{8} = \left(\frac{1}{2}\right)^3 \rightarrow \text{Decays to half for 3 times in 24 days} \rightarrow t_{1/2} = \frac{24 \text{ days}}{3} = 8 \text{ days}$$

5. How much energy is lost or gained when 1 mol of cobalt-60 undergoes beta decay?

The mass of a $^{60}_{27}\text{Co}$ atom is 59.933819 g/mol, that of a $^{60}_{28}\text{Ni}$ atom is 59.930788 g/mol, and that of an electron is 5.4858×10^{-4} g/mol.

$$\Delta m = 59.933819 + (59.930788 + 0.000549) \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}} \cdot (3 \times 10^8 \text{ m/s})^2 = 2.234 \times 10^{-11} \text{ J (lost)}$$

6. Draw and name the structural isomers of heptane.

